

Post-Plant Geometry: A Mirage Survival Analysis of Tactical Positioning in Counter-Strike 2

Other Sports
Paper ID: 109

1. Introduction

Over the last decade, electronic sports, or esports, have grown from niche entertainment into a global industry with viewership and financial stakes comparable to many traditional sports [1]. Counter-Strike 2 (CS2) is one of the flagship titles in this space. In CS2, two five-player teams alternate between attacking and defending. The attacking side (Terrorists, Ts) must plant a bomb at one of two bomb sites within a one minute and fifty second timer, while the defending side (Counter-Terrorists, CTs) must prevent the plant or defuse the bomb once it is armed. Rounds end when the bomb explodes, is successfully defused, one team is eliminated, or time runs out. The first team to win 13 rounds wins the match. These simple rules support a deep tactical game where team coordination, map control, and second-by-second decision making determine success.

Within this structure, some of the most decisive and memorable moments occur after the bomb has been planted, forcing the defending team into what is known as a 'retake'—reclaiming the bomb site under extreme time pressure before detonation. On maps like Mirage, post-plant scenarios create tactical chaos: defenders scramble through tight chokepoints while attackers hold elevated positions with overlapping angles of fire. One defender might be trapped on-site with depleted utility, forced to hold an exposed angle. Another posts at long range behind temporary cover, hoping enemies push predictably. A third rotates on a delayed flank—a high-risk gamble that could swing the round or arrive too late to matter. All three players are technically in the same "retake," yet they face radically different survival probabilities based on positioning, utility state, and timing. Yet current CS2 analytics typically describe these situations with coarse aggregates such as overall retake win rate or 1vX clutch percentage. These statistics treat all post-plants as interchangeable and largely ignore where players are standing, how much time remains on the bomb, which lines of sight are blocked by utility, and how many angles are about to open as smokes and molotovs expire.

This work is motivated by that gap. Rather than viewing clutches and retakes as fixed odds like "this team wins 40% of A-site retakes" or "1v2s succeed 29% of the time," it asks how space, time, and resources interact with the geometry of the map to shape survival in post-plant situations on Mirage. In practical terms, the central question is: given a player's position, the angles they are holding or exposed to, the available utility, and the remaining bomb time, what makes one post-plant setup robust and another a death sentence, and how does that answer change as utility dissipates and sightlines open?

1.1. Analytical Framework

We address this gap through Cox proportional hazards models applied to episode-level positioning data from 569 professional matches spanning 2023-2025. Rather than treating retakes as binary

outcomes, we decompose each retake into player-episodes: continuous periods in specific map positions. The framework divides Mirage into tactical zones and tracks survival as players move between positions during retakes. This granularity enables us to capture how survival probability changes as utility expires, teammates are eliminated, and the bomb timer depletes.

Using grouped cross-validation and cluster-robust standard errors to account for correlated observations, our models achieve concordance indices ranging from 0.748 to 0.806 (cross-validation), enabling position-specific risk quantification across team-site combinations. We partition the map into 291 discrete spatial zones using custom polygons (Figure 2), with each zone representing a tactically distinct position. The models incorporate time-varying covariates including utility usage, weapon loadouts, numerical advantage, teammate support, and empirically-derived visibility relationships—all measured at episode start to avoid information leakage. Critically, all spatial relationships (sightlines, connectivity, crossfire potential) are derived empirically from professional play rather than assumed from map geometry. The analysis reveals, for instance, that as a T-side player, holding Jungle Cubby (Zone 154) reduces elimination hazard by 7% ($HR=0.93$, $p<0.01$), while having numerical superiority and teammate support reduces individual risk by 6-8% ($HR=0.938$ for advantage, $HR=0.985$ for teammates, $p<0.001$)—illustrating how spatial positioning and numerical advantage combine to create distinct survival profiles.

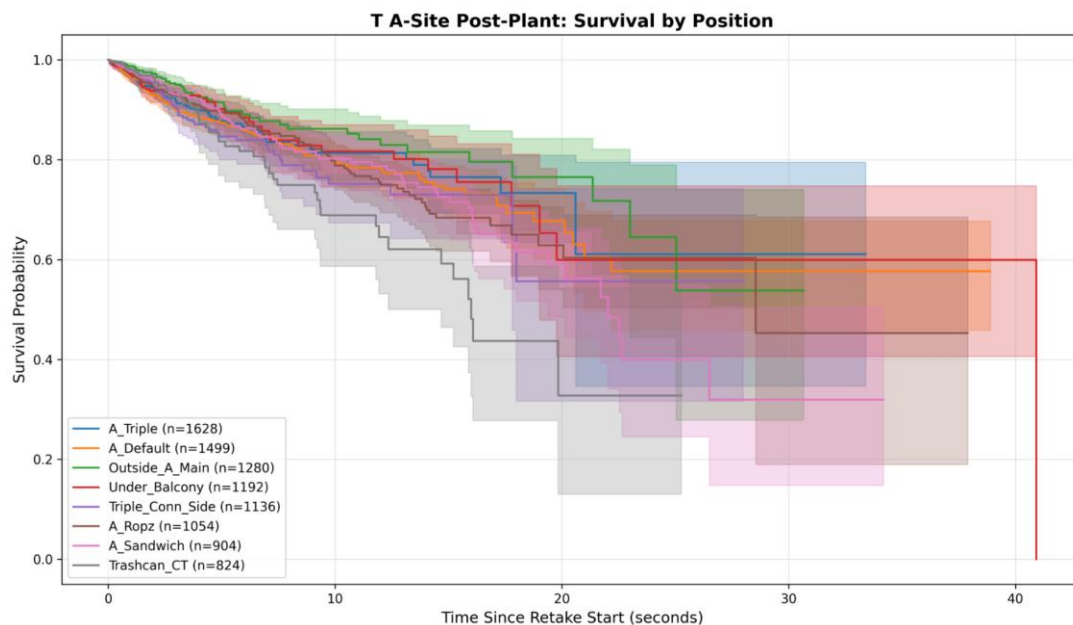


Figure 1: Kaplan-Meier survival curves for T-side players at A-site by initial position, illustrating position-dependent survival profiles. Outside A Main (green, Zone 62) shows superior survival probability (~80% at 20 seconds) compared to exposed positions like Trashcan CT (gray, Zone 8 ~45% at 20 seconds). Shaded regions indicate 95% confidence intervals. Our Cox proportional hazards models adjust for equipment, crossfire support, and numerical advantage to quantify position-specific risk while controlling for these confounding variables.

This framework provides teams, analysts, and players with actionable insights into post-plant positioning strategy, quantifying the survival impact of tactical decisions that are currently evaluated through intuition alone. By revealing how position, utility, and timing interact to shape elimination risk, the analysis enables data-driven optimization of retake setups and individual positioning choices. The position-specific hazard ratios and survival curves produced by our models can inform coaching decisions, pre-round planning, and post-match review, offering competitive advantages. This paper is organized as follows: Section 2 describes our data collection, zone classification, and episode tracking methodology. Section 3 presents the Cox proportional hazards modeling approach and validation strategy. Section 4 reports model performance and position-specific survival patterns across utility states, numerical advantages, and crossfire configurations. Section 5 discusses tactical implications, challenges conventional assumptions, and identifies information state as a critical limitation for future work.

2. Dataset Creation

A key advantage of analyzing Counter-Strike 2 is the availability of high-fidelity match replay files. Unlike traditional sports where tracking data requires specialized camera systems or manual annotation, CS2 demo files capture complete match state at the game's native subtick architecture, including player positions, weapon inventories, utility usage, and all damage events. This granular data enables precise reconstruction of tactical scenarios without the measurement error inherent in computer vision-based tracking systems.

2.1. Demo Selection

We selected the map Mirage for analysis due to its consistent popularity in professional play and structural stability—the map has received no geometry changes since Counter-Strike 2's release in October 2023, ensuring tactical patterns reflect current meta rather than adaptation to map updates. This single-map focus controls for map-specific geometry and strategic conventions while providing sufficient sample size for robust statistical analysis.

Our dataset comprises demos from the first CS2 professional tournament, IEM Melbourne 2023 (October 16, 2023), through PGL Masters Bucharest 2025 (October 26, 2025). To ensure competitive quality and tactical rigor, we applied two filters: minimum prize pool of \$100,000 USD and LAN (Local Area Network)-only events¹. These criteria naturally select tier-1² and tier-1.5³

¹LAN (Local Area Network)-only events: Tournaments where all players compete in-person at a physical venue, eliminating online latency and ensuring maximum competitive integrity compared to online matches.

²Tier-1: Teams that consistently compete in major international LAN tournaments, representing the highest level of professional Counter-Strike competition.

³Tier-1.5: Teams that primarily compete in large online tournaments and qualifiers for LAN tournaments who make occasional appearances at larger LAN events, representing high-level but not top-tier professional play.

teams where tactical execution and positioning decisions represent elite-level play. All demos were sourced from HLTV's⁴ public archive.

After filtering for data quality, our final dataset contains 569 matches across 72 tournaments. Major tournaments represented include CS Asia Championships 2023, ESL Impact League Season 4 Finals, IEM Katowice 2024, PGL CS2 Major Copenhagen 2024, YaLLa Compass 2024, IEM Rio 2024, Perfect World Shanghai Major 2024, BLAST Rivals 2025 Season 1, IEM Dallas 2025, ESL Pro League Season 22, and more.

2.2. Zone Classification

To capture the spatial complexity of post-plant positioning, we manually defined 291 tactical zones across Mirage's two bombsites and connecting areas. Unlike prior work that aggregates player positions into coarse regions (e.g., "A Site" or "CT Spawn"), our zone definitions reflect granular tactical distinctions—separating high-traffic chokepoints, common defensive positions, off-angles,⁵ anchor positions,⁶ and utility-dependent angles that exhibit distinct survival dynamics. The construction of these polygon zones was assisted by Dexter, a professional CS2 player with years of experience as an In-Game Leader (IGL)⁷ in the Tier 1 ecosystem, whose competitive expertise informed zone boundaries accounting for plant positions, defensive setups, and common engagement angles.

Zone coordinates were defined using a private CS2 server, allowing precise boundary mapping through in-game positioning and coordinate extraction. Each zone is stored as a three-dimensional polygon in GeoJSON format with x, y, and z coordinates corresponding to the game's world space. Player positions from demo files are assigned to zones via point-in-polygon testing at each game tick (64Hz). This tick-level classification enables precise tracking of player movement across zones, capturing rapid repositioning during retake scenarios. For overlapping or nested zones—such as specific cover positions within larger site areas—players are assigned to the smallest enclosing zone to maximize spatial granularity. This approach ensures that tactical nuances, such as holding tight angles versus open positions within the same bombsite, are distinguished in the survival analysis.

⁴HLTV website: <https://www.hltv.org/>

⁵Off-angles: An unconventional or unexpected defensive position that deviates from standard holding spots, designed to catch attackers by surprise.

⁶Anchor positions: A primary defensive position held by a player responsible for initially defending a bombsite, typically offering strong multi-kill potential and sustained engagement potential.

⁷ In-Game Leader (IGL): The strategic captain of a Counter-Strike team who makes tactical decisions during matches, including site execution plans, utility usage, defensive setups, and mid-round adjustments. IGLs possess deep understanding of map control, timing, and positional play.

For example, on A-site, we distinguish between macro positions such as "Firebox" (Zone 32), "Default A" (Zone 30), "Palace" (Zone 74), and "A Main" (Zone 62), as well as micro positions like "Ropz" (Zone 14), "Jungle Cubby" (Zone 154), "Connector/Jungle Cubby" (Zone 47), and "Trash Can" (Zone 8). Similarly, B-site zones separate macro areas including "Default B" (Zone 228), "Van Tucked" (Zone 257), "Bench" (Zone 246), and "Apps End" (Zone 262), along with micro positions such as "Pillar 1" (Zone 250), "Dumpster" (Zone 184), "Get Right" (Zone 213), and "Market Exit" (Zone 182). This level of granularity contrasts sharply with prior work that might collapse all A-site positions into a single "Bombsite A" category, losing critical information about exposure risk, rotation paths, and sightline availability. Figures 2 and 3 illustrate the complete zone classification for both bombsites.

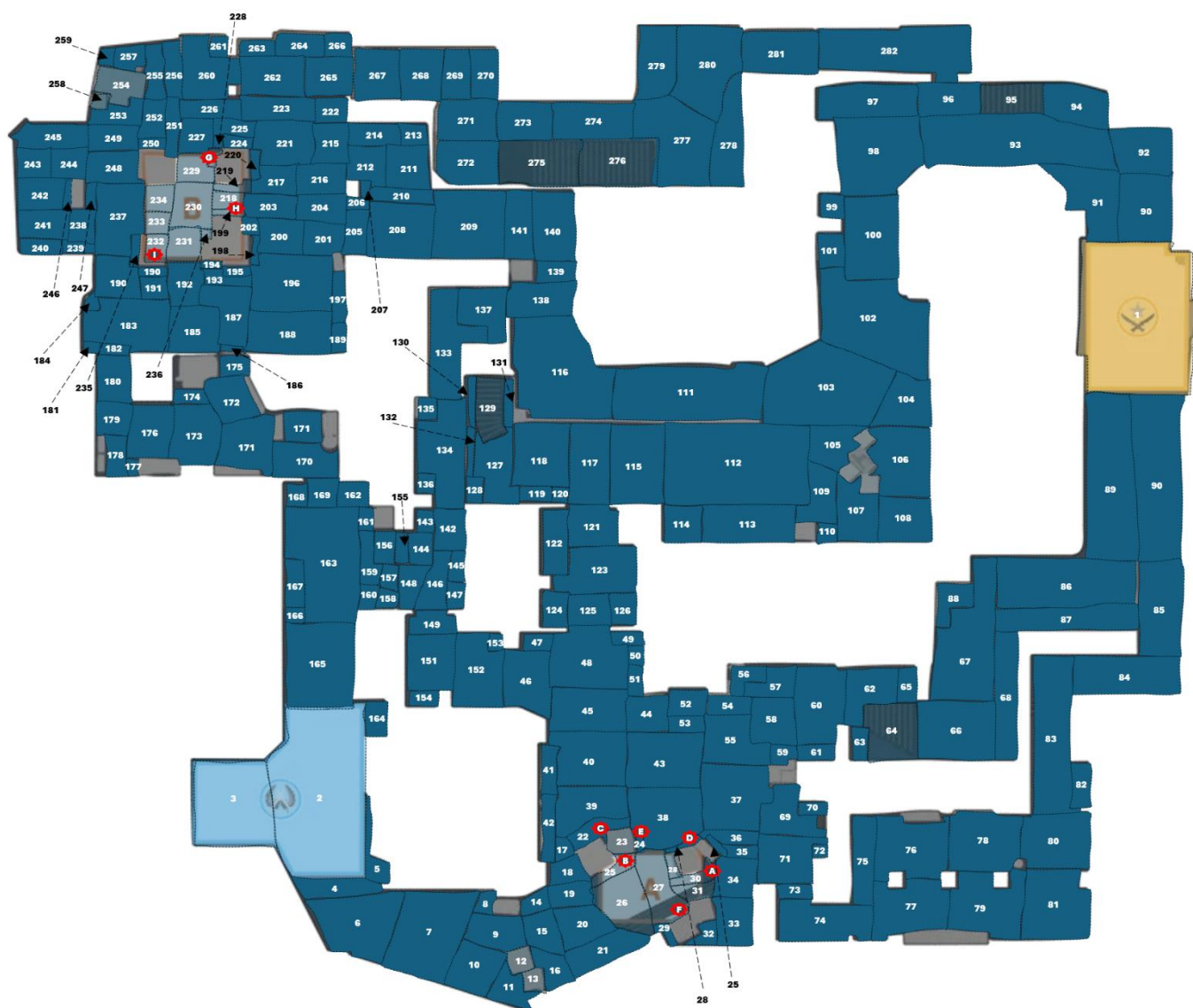


Figure 2: Mirage Upper Level Zone Classification. Custom tactical zone boundaries showing numbered polygons representing distinct tactical positions defined from professional positioning patterns. Zones group positions with similar sightline exposure and engagement risks. Transparent zones indicate bombsite boundaries for spatial reference, while blue polygons represent all tactical positions including defensive holds, rotation corridors, and attacking entry points. Red markers denote common plant positions. Zone numbers correspond to callouts referenced in the analysis (e.g., Zone 14 = Ropz, Zone 30 = Default A, Zone 154 = Jungle Cubby). A complete mapping of all 291 zones is provided in Appendix B.

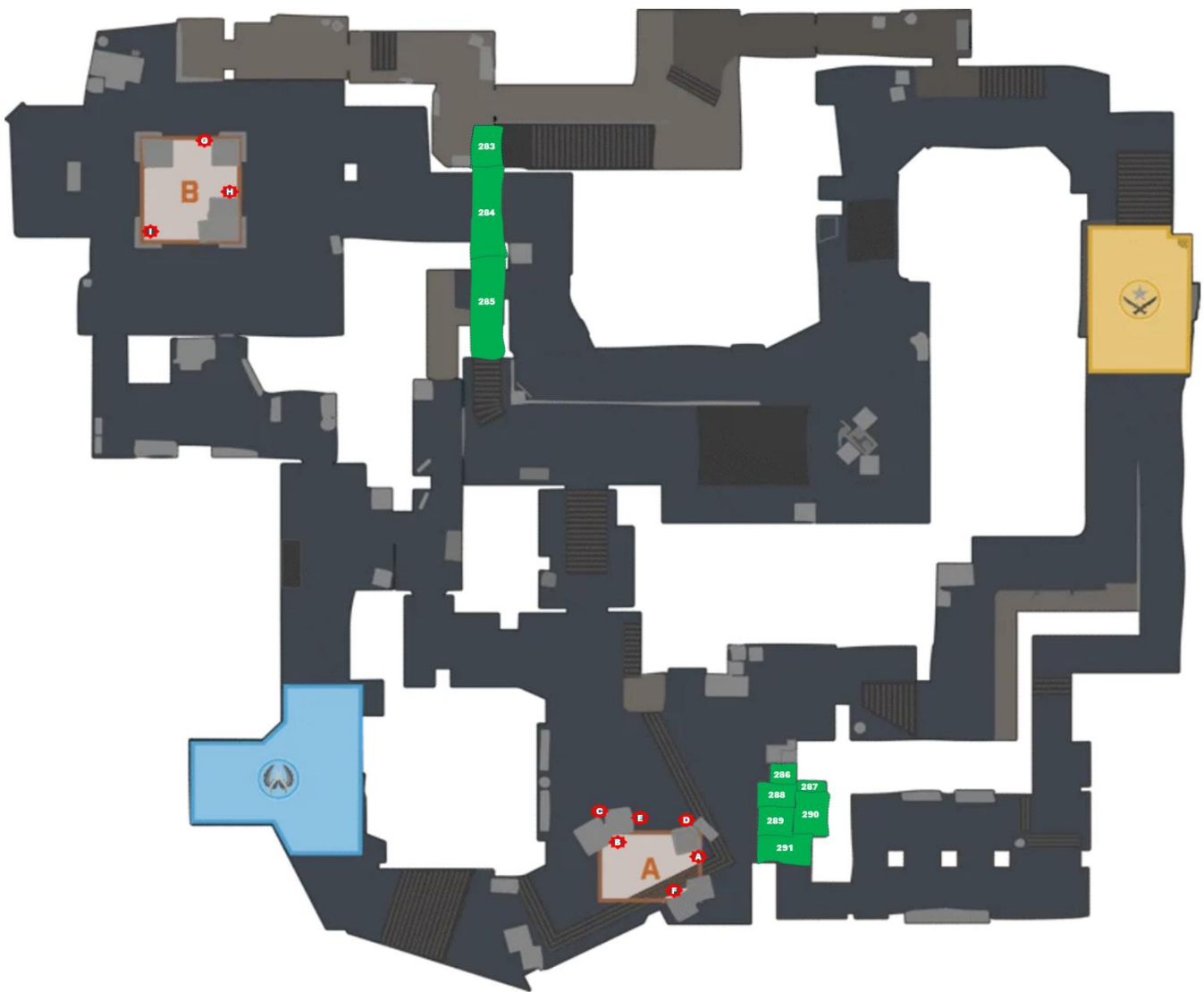


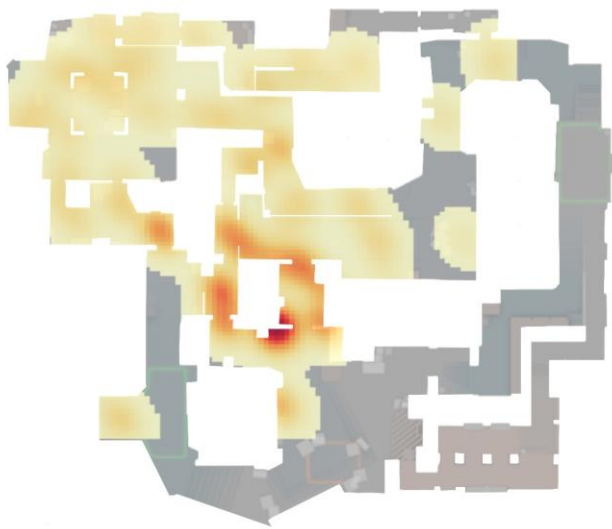
Figure 3: Mirage Lower Level Zone Classification. Custom tactical zone boundaries for Mirage's lower level, showing numbered polygons representing distinct tactical positions. Green zones indicate lower-elevation areas, while red markers denote common plant positions. Zone numbers correspond to specific callouts referenced in the analysis (e.g., Zone 287 = Under Balcony Tucked). A complete mapping of all 291 zones is provided in Appendix B.

2.3. Retake Detection

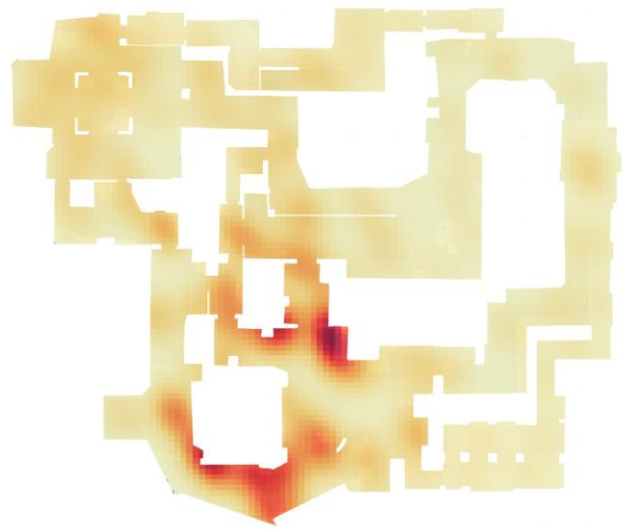
Identifying retake scenarios requires first detecting when and where the bomb is planted, then classifying if a retake is occurring. Plant detection begins with awpy's built-in event parser, which extracts the plant tick (64 Hz timestamp), planter identity, and location from demo tick data. When awpy fails to detect a plant—typically due to parsing inconsistencies in early CS2 demos or mid-round disconnections—we apply a fallback heuristic: if the round ends in a bomb defusal, we infer the plant tick by walking backward from the explosion/defusal event using the known 40-second C4 timer (2,560 ticks at 64 Hz).

Retake engagement classification uses movement trajectory analysis to distinguish active retakes from economic saves and exit-frag hunting. We compute zone-to-zone path vectors for each surviving CT following 20 seconds pre-plant, comparing their movement patterns against typical retake and save trajectories. CTs moving toward bombsite zones (Figure 4) who subsequently attempt to defuse or engage enemies while positioned on-site are classified as retaking. CTs moving away from the bombsite without on-site combat engagement are classified as saving and excluded. Similarly, CTs who engage in combat near the bombsite perimeter but make no sustained push into bombsite zones—seeking economic damage rather than retake commitment—are excluded as exit-frag behavior. Rounds ending in defusal are automatically classified as retake attempts regardless of individual player pathing, as defusal confirms at least one CT reached the bombsite under pressure. This spatial filtering leverages the granular zone definitions to capture genuine tactical repositioning under fire rather than passive rotation or opportunistic damage trading. Bombsite classification (A vs. B) relies on the planter's XYZ coordinates at the plant tick, matched against the zone polygons defined in Section 2.2. In rare cases where the plant occurs outside defined zones—such as on elevated objects or map geometry edges—we assign the bombsite using nearest-neighbor distance to A-site and B-site zone centroids.

This pipeline successfully identified 3,508 retake scenarios from the 569 tournament matches, comprising 2,297 A-site retakes and 1,211 B-site retakes.



(a)



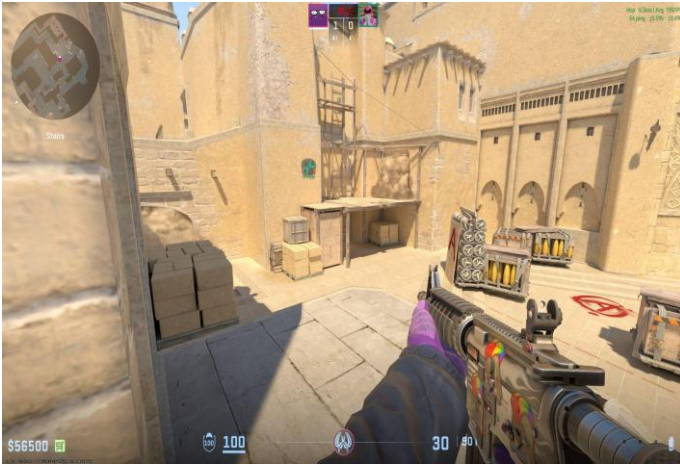
(b)

Figure 4: CT positioning heatmaps during A-site post-plant scenarios (3,508 retakes). (a) Save positioning shows CTs retreating from the A-site through Mid Connector (Zone 123) to Underpass (Zone 285) and via Jungle (Zone 152) through Market Entry (Zone 169), preserving weapons rather than engaging. (b) Retake positioning shows CTs converging toward the bombsite from multiple entry points: CT Spawn (Zone 2) through Ticket (Zone 11), Window Room (Zone 148) and Jungle (Zone 152) via Connector (Zones 121, 125) to Stairs Platform (Zone 44), and direct pushes through Window (Zone 134). Heat intensity reflects position frequency across all scenarios.

2.4. Episode Tracking

Traditional survival analysis treats each player's retake as a single observation, measuring time from bomb plant to death or round end. However, this coarse representation discards crucial temporal dynamics: mortality risk varies dramatically as players traverse different zones with changing tactical conditions. A Counter-Terrorist might spend five seconds safely repositioning behind a smoke, three seconds in a connecting corridor, then die within two seconds upon entering

an exposed platform facing the bombsite. Treating this scenario as one eight-second survival observation obscures the zone-specific risk profiles that define tactical decision-making.



(a)



(b)

Figure 5: (a) Counter-Terrorist engaging in A-site retake at Stairs Platform (Zone 44, HR=1.070 from baseline) with active smoke coverage blocking sightlines from A Main (Zone 62) and Palace (Zone 74), resulting in empirically-observed hazard rate of $\lambda \approx 0.065$ deaths/sec (mean survival 3.0 seconds with smoke). (b) Same spatial position after smoke dissipation exposes Stairs Platform to unobstructed Palace sightlines, yielding elevated hazard from combined zone risk and loss of smoke protection (observed mean survival 0.5 seconds post-dissipation). Traditional survival analysis treating this as a single observation would collapse the discrete hazard shift at smoke expiration, obscuring the time-varying covariate effect critical to Cox model specification.

To capture time-varying hazard rates, we employ the counting process formulation of Cox regression [3], widely used in medical research for modeling patients transitioning through hospital wards with differing mortality risks. This [start, stop] interval representation decomposes each retake into multiple episodes—discrete periods where a player occupies a single zone under specific covariate conditions. Each episode contributes an independent observation to the Cox model: the hazard function is re-evaluated at each zone transition, allowing player state (health, utility coverage, numerical advantage, etc.) and environmental factors (sightline exposure, connectivity) to dynamically influence survival probability as the retake unfolds.

Our implementation defines episodes by zone occupancy: an episode begins when a player enters a zone and ends when they either transition to a different zone (censored observation) or die (event). From the 3,508 retake scenarios identified in Section 2.3, we extracted 65,688 episodes across four model configurations: 16,928 CT episodes on A-site (1,191 deaths), 7,897 CT episodes on B-site (637 deaths), 28,094 Terrorist episodes on A-site (1,558 deaths), and 18,748 Terrorist episodes on B-site (754 deaths). Episode counts reflect both the distribution of retakes across bombsites (66%

A-site, 34% B-site) and asymmetric tactical roles: Counter-Terrorists executing coordinated site pushes reposition more frequently across approach routes and entry points (averaging 10.1 episodes per A-site retake) compared to Terrorists defending the planted bomb from established positions (7.4 episodes per A-site retake). Each episode is characterized by its spatial context (zone identity, bombsite), temporal extent (entry tick, exit tick, duration), outcome (binary death indicator), and scenario linkage (demo file, round number, plant tick) enabling within-retake clustering in our Cox models.

Most episodes are right-censored: approximately 93% of CT episodes and 95% of T episodes end with the player transitioning to a different zone rather than dying. These censored observations remain statistically valuable because they establish lower bounds on survival time under specific conditions. When a Counter-Terrorist survives 5 seconds in a connecting corridor with active smoke coverage before advancing to the next zone, this censored episode informs the Cox model that the hazard rate for "corridor + smoke" was sufficiently low to permit at least 5 seconds of survival. Censored episodes contribute to the risk set—the pool of individuals still alive and observable at each death time—enabling the partial likelihood estimation to compare covariate profiles between those who died and those who survived in similar tactical situations. Only when an episode ends in death (event indicator = 1) does it contribute a death event to the likelihood function; however, all episodes, censored or not, shape the baseline hazard estimation through their duration and covariate values.

Each episode record captures spatial, temporal, and outcome data: zone identifier, bombsite designation, entry and exit ticks (64 ticks $\approx$ 1 second), duration, and binary death indicator. Episodes are linked to their parent retake scenario via demo file, round number, and plant tick. Episode-level covariates include player state variables (health points, armor status, weapon type, defuse kit possession) and dynamic environmental factors detailed in Sections 2.5-2.6: empirically-derived visibility to opponents, connectivity to the bombsite, numerical advantage, and utility coverage (smoke/molotov time remaining). Because episodes from the same retake scenario are inherently correlated (shared opponents, shared utility usage, shared round economy), we employ robust variance estimation with clustering by retake identifier [4]. This adjustment accounts for within-retake dependence in the standard errors, ensuring valid statistical inference despite the hierarchical data structure.

2.5. Visibility and Connectivity

Mortality risk in post-plant scenarios depends critically on spatial relationships: which zones expose players to enemy sightlines, and how quickly teammates can traverse the map to provide support. Rather than imposing geometric assumptions through raytracing algorithms or manual map annotation, we derive visibility and connectivity relationships empirically from observed gameplay. This data-driven approach captures tactical realities that geometric models miss—head glitch positions, common holding angles, boost spots, and movement shortcuts employed by professional players—ensuring our spatial covariates reflect actual competitive behavior rather than idealized map geometry.

We construct a visibility graph by analyzing weapon damage events across all retake scenarios. When player A in zone Z_1 deals damage to player B in zone Z_2 using a weapon requiring line-of-sight (rifles, pistols, AWP), we record a directed edge $Z_1 \rightarrow Z_2$ in the visibility graph. Aggregating

damage events across our retake scenarios yields an empirical sightline network: each edge represents a zone pair where direct visual contact occurred during actual professional play. For each episode, we compute visibility exposure as the count of occupied enemy zones with confirmed sightlines to the player's current zone. This metric quantifies tactical vulnerability: positions exposed to multiple enemy angles exhibit higher mortality hazard than those visible from only a single direction (see Figure 6). Unlike geometric raytracing, this approach inherently captures effective sightlines—angles where players actually engage—excluding technically-visible positions that professional players avoid due to exposure disadvantage.



Figure 6: Empirical visibility exposure on B-site. Counter-Terrorist positioned near Dumpster (Zone 184) during B-site retake with visibility to multiple enemy-occupied zones. The visualization shows confirmed damage-based sightlines to two Terrorist players: one at Van Jump Up (Zone 259) and another at Pillar 1 (Zone 250). In this tactical scenario, the CT faces visibility exposure from 2 enemy-occupied zones. This metric quantifies tactical vulnerability: positions exposed to multiple enemy angles exhibit higher mortality hazard in Cox models. Sightline edges are derived empirically from retake scenarios, capturing effective engagement angles used in professional play rather than geometric line-of-sight calculations.

Complementing the visibility graph, we construct a connectivity graph by tracking player zone transitions across all retake scenarios (Figure 4b). When a player moves from zone Z_1 to zone Z_2 , we record the transition time (in game ticks) between entry into Z_1 and entry into Z_2 . Aggregating

these observations yields a weighted directed graph where edge weights represent the median travel time between adjacent zones. This empirical connectivity network captures realistic movement patterns: professional players exploit optimal routes, jump shortcuts, and boost positions that reduce transit times beyond naive walking-speed estimates. For each episode, we compute connectivity to bombsite as the shortest-path travel time from the player's current zone to the planted bomb location using these empirical edge weights. This covariate quantifies reinforcement potential: a defending Terrorist positioned far from the bombsite (high connectivity value) faces elevated mortality risk when isolated, while teammates in zones with low connectivity values can rapidly rotate to provide support or trade kills.

For a player in zone z at time t , connectivity to adjacent zones is computed as:

$$c(z) = \min_{paths\ p} \sum_{e \in p} w(e) \quad (1)$$

Where $w(e)$ represents the empirical medial travel time for edge e .

The resulting spatial graphs provide comprehensive coverage of tactical positioning across Mirage: the visibility graph contains sightline data for 291 zones across the map, while the connectivity graph captures movement patterns between all accessible zone pairs. To validate these empirical networks, we verified that visibility edges align with known strategic positions (e.g., Default plant position on the A bomb site (Zone 30) has confirmed sightlines to multiple common defensive zones like Ticket Entry (Zone 9), Triple CT Side (Zone 18), and Ninja (Zone 29). Connectivity values reflect domain knowledge about map traversal times (e.g., Cat (Zone 116) to Stairs Platform (Zone 44) averages 6 seconds via optimal routes). These graphs remain static across all episodes within our dataset, serving as foundational spatial covariates that interact dynamically with time-varying factors such as utility coverage, numerical advantage, and player health to determine episode-specific mortality hazard.

2.6. Covariate Extraction

Beyond spatial context (visibility and connectivity from Section 2.5), each episode is characterized by a rich set of time-varying covariates capturing player state and tactical conditions. We organize these covariates into four categories: (1) player resources (health, armor, weapon loadout, equipment, kit status), (2) utility state (smoke and molotov coverage with time-remaining calculations), (3) numerical advantage (alive teammate and opponent counts), (4) bomb plant location, (5) Crossfire detection and (6) Team composition. All covariates are extracted at episode initiation—the tick when a player enters a zone—but only for rounds identified as retake scenarios after bomb plant. This ensures that hazard function evaluations reflect the tactical conditions present at the start of each risk period while limiting analysis to strategically relevant post-plant situations. Covariates remain constant within an episode but update at each zone transition, allowing the model to estimate how changing tactical conditions influence mortality risk as retakes unfold.

2.6.1. Player Resource Covariates

Player resource covariates quantify combat readiness at episode entry. Health points (HP, range 1-100) and armor status (binary indicators for kevlar vest and helmet) directly influence survivability in gunfights. Weapon type is encoded as a categorical variable with granular distinctions: primary rifles are separated by model (AK-47, M4A4, M4A1-S) to capture damage and spray pattern differences; high-impact pistols (Desert Eagle, Five-Seven, Tec-9) are isolated due to their armor-penetration capabilities; other pistols and SMGs are grouped by class; and the AWP sniper rifle and shotguns form distinct categories reflecting specialized engagement patterns. For Counter-Terrorists, defuse kit status is tracked via two binary indicators: `has kit` (currently possesses a defuse kit) and `picked_up_kit` (acquired a defuse kit during the retake, typically from a fallen teammate). The `picked_up_kit` variable proves particularly informative: acquiring a defuse kit mid-retake signals increased retake ability, as defuse kit possession reduces bomb defusal time from 10 seconds to 5 seconds—often decisive in time-constrained post-plant scenarios. Equipment counts (flashbangs, HE grenades, smoke grenades, molotovs/incendiaries remaining) complete the resource profile, though active utility effects are tracked separately via time-remaining calculations detailed below.

2.6.2. Utility State Covariates

Utility state covariates capture the tactical impact of grenades that persist in the environment beyond their deployment. CS2 demo files record each utility activation with a spawn tick and expiration tick, enabling precise calculation of remaining duration at any point during the retake. For each episode, we compute `smoke_time_remaining` and `molly_time_remaining` as the seconds until utility expiration in the player's current zone at episode entry. These continuous covariates (range 0-18 seconds for smokes, 0-7 seconds for molotovs) quantify protective coverage: fresh utility (high time-remaining values) blocks sightlines or denies area access, while dissipating utility (low values) signals imminent exposure. We additionally encode binary indicators for `fresh_smoke` (≥ 10 seconds remaining) and `fresh_molly` (≥ 3 seconds remaining) to capture threshold effects where utility transitions from tactically effective to negligible. Flashbang effects are tracked via a binary indicator (`flashed`) marking vision impairment during the episode, as flashbang duration (~2-5 seconds) varies with distance and line-of-sight exposure. As demonstrated in Figure 5, the discrete hazard shift when smoke coverage expires exemplifies why time-remaining covariates prove critical for accurate mortality risk estimation.

2.6.3. Numerical Advantage Covariates

Numerical advantage covariates quantify the balance of forces at episode initiation. For each player, we compute `num_allies` and `num_enemies` as the count of living teammates and opponents at episode entry, with the difference `numerical_advantage = num_allies - num_enemies` serving as a summary metric (positive values indicate numerical superiority, negative values indicate disadvantage). These covariates prove critical for mortality risk estimation: a Counter-Terrorist executing a 5v2 retake faces fundamentally different hazards than one in a 2v4 scenario, even when occupying the same zone with identical equipment. Team composition effects extend beyond simple headcount: we track whether teammates occupy zones with confirmed sightlines to the player's position (potential trading partners) and compute aggregate team resources (total HP, utility counts) to capture coordinated tactical strength. The interplay between numerical advantage, spatial positioning (visibility/connectivity from Section 2.5), and utility coverage creates

a high-dimensional covariate space where Cox models can estimate context-specific mortality hazards across the diverse tactical situations arising in professional retake attempts.

2.6.4. Bomb Plant Location

Bomb plant location is encoded as a categorical variable distinguishing common plant positions identified from professional play (see Figure 2 for classified plant zones). Plant location affects both Terrorist defensive positioning—proximity to the bomb influences post-plant hold angles and rotation options—and Counter-Terrorist tactical constraints, including defuse exposure and the angles from which retaking players can safely approach the planted bomb. This spatial covariate interacts with zone-level visibility and connectivity metrics to capture how plant position shapes the tactical geometry of retake scenarios.

2.6.5. Crossfire Detection

Crossfire⁸ setup and trading potential (securing retaliatory eliminations after a teammate dies) quantify defensive coordination when teammates occupy positions that enable mutual support. In a post-plant, crossfire exists when two or more teammates satisfy either: (a) both have direct line-of-sight visibility to the same enemy-occupied zone (immediate trading ability if one defender is eliminated), or (b) one teammate has visibility while another occupies an adjacent zone within one connectivity-graph transition, enabling rapid rotation to establish sightlines (swing⁹ potential). Using the empirical visibility graph (Section 2.5) and connectivity graph (Figure 4), we detect crossfire configurations by identifying episodes where a defender's current zone either shares visibility edges with teammate-occupied zones targeting the same enemy position, or connects via single-transition paths to zones with such visibility. The binary covariate `has_crossfire_support` flags episodes where defenders benefit from coordinated positioning, capturing the protective effect of teammate backup: a defender holding an angle with crossfire support faces lower mortality risk than one isolated without trading partners, as crossfires enable trading opportunities and discourage enemy aggression. Figure 7 illustrates this detection: panel (a) shows two defenders positioned at Triple and Palace, where the Palace player remains hidden but can swing out to Heaven to establish a crossfire, while panel (b) demonstrates the Palace player actively swinging to create overlapping angles of fire with the Triple teammate. This highlights how the model identifies

⁸Crossfire: A defensive setup where multiple players cover overlapping angles, enabling trade kills if one is eliminated.

⁹Swing: In tactical FPS terminology, “swinging” refers to rapidly transitioning from cover into an engagement angle, typically by moving from a concealed position into a sightline.

crossfire potential not only from static sightlines but also from positions that enable rapid swing movements to create mutual support.



(a)



(b)

Figure 7: Crossfire detection on Mirage B-site: (a) Palace player hidden with crossfire potential via swing to Heaven, (b) Palace player actively swung to Heaven creating overlapping angles with Triple defender. Solid lines indicate direct sightlines; dashed lines show swing movement capability.

2.6.6. Teammate Composition

Beyond teammate headcount, we capture team quality through aggregated features measuring the effectiveness of available backup. For each player-episode, we compute statistics across all living teammates at the retake's onset: mean normalized health (0-1), mean normalized armor (0-1), helmet ownership rate, and average utility count (grenades). These features distinguish teammates with full combat effectiveness from damaged players on economic rounds. We count teammates by weapon category (rifles, pistols, SMGs, AWP) to capture firepower capability, and track how many are currently impaired by utility (in smoke, molly, or flashed). Rather than identifying exact teammate zones—which resulted in sparse binary indicators with extremely low occurrence rates (<1%) causing complete separation issues—we leverage the zone connectivity graph (Figure 4) to cluster teammate positions into meaningful tactical groupings. This aggregation approach captures overall team quality (average health, armor, helmet rate, utility availability) and tactical coordination (crossfire support, numerical advantage) while avoiding convergence failures from rare positional combinations. The model thus learns whether a player has effective backup (healthy, well-armed teammates in connected zones providing crossfire support) versus mere numerical presence. For Counter-Terrorist retakes, we apply the same teammate composition features as Terrorist models, with the addition of defuse kit indicators: both the player's kit ownership and the kit rate among teammates, since kits reduce defuse time from 10 to 5 seconds—

a critical advantage in post-plant scenarios. All teammate features derive from baseline positions at plant time to prevent information leakage from future movements.

2.7. Data Processing Pipeline

Raw demo files (.dem format) were parsed using a combination of demoparser2 and awpy Python libraries. The demoparser2 library interfaces with Valve's official demo parsing tools to extract high-volume game state information at tick-level granularity (64 ticks per second), while awpy provides utility functions for tactical feature extraction and spatial analysis. Rather than iterating through millions of ticks per match, the pipeline employs targeted extraction: we first identify bomb plant events and subsequent retake scenarios (Section 2.3), then extract detailed game state beginning 20 seconds prior to plant. This pre-plant buffer captures utility deployments (smokes, molotovs) thrown before the plant that remain active during early retake episodes, ensuring complete utility state tracking from retake initiation. The parsing pipeline then implements zone classification (2.2), episode decomposition (2.4), visibility and connectivity graph construction (2.5), and covariate extraction (2.6) for the identified retake windows.

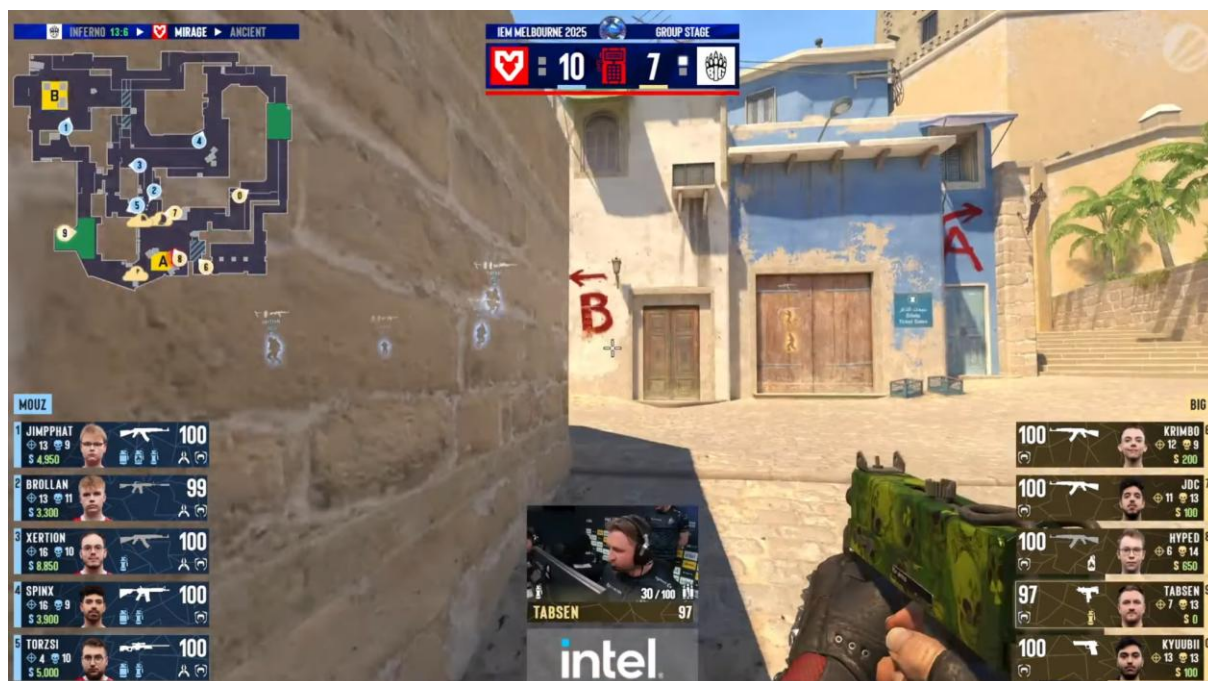


Figure 8: Comprehensive covariate snapshot at retake episode initiation. Professional match frame (IEM Melbourne 2025, Mouz vs. BIG) showing A-site retake immediately after bomb plant on Default A (Zone 30). Player HUD displays resource states: Tabsen in Deep CT Spawn (Zone 3) with 97 HP, kevlar+helmet, MP9, and flashbang remaining. Multiple zone-specific smokes provide coverage with varying durations (Ticket/Zone 9: 16s, Stairs Platform/Zone 44: 7s, Jungle Open 1A/Zone 45: 14s). Crossfire detection identifies defensive positioning from JDC (Zone 52), Hyped (Zone 30), and Krimbo (Zone 74). The Cox model ingests this complete tactical context—spatial positioning, player resources, utility states, numerical advantage, crossfire support—at each

episode transition, updating dynamically as conditions evolve. Source: *IEM Melbourne 2025, Mouz vs. BIG*.

Figure 8 exemplifies the dense tactical information captured at each episode transition: spatial positioning (zone identifiers), player resources (HP, armor, weapons, equipment), utility states (smoke/molotov time remaining), team composition (weapon loadouts, numerical advantage), and crossfire configurations. Processed data is serialized to Apache Parquet format, yielding `ct_episodes.parquet` (24,825 episodes) and `t_episodes.parquet` (46,842 episodes), each containing 44-46 base feature columns capturing these tactical dimensions. After Cox model preparation—including zone and weapon dummy variable encoding and interaction term construction—the final feature matrices expand to 62-120 covariates depending on model specification and bombsite topology. Total dataset size is approximately 45 MB compressed, enabling rapid iteration during model development. Table 1 summarizes the final dataset composition by team and bombsite.

Table 1: Dataset composition by team and bombsite.

Team	Bombsite	Episodes	Deaths	Censored	Event Rate	Mean Duration (s)	Unique Retakes
CT	A	15,336	1,137	14,199	7.4%	1.9	2,297
CT	B	6,024	576	5,448	9.6%	1.9	1,211
CT Total	Both	21,360	1,713	19,647	8.0%	1.9	3,508
T	A	26,882	1,522	25,360	5.7%	2.8	2,297
T	B	17,446	714	16,732	4.1%	2.8	1,211
T Total	Both	44,328	2,236	42,092	5.0%	2.8	3,508
Overall	Both	65,688	3,949	61,739	6.0%	2.5	4,621

Note: Episodes are censored at zone transition or round end. Event rate = deaths/episodes. Mean duration measures time (in seconds) from bomb plant to death or censoring event. Each retake scenario generates episodes for both teams simultaneously (one episode per living player). The 3,508 unique retakes yield 65,688 total player episodes across both teams. Analysis sample excludes episodes with missing covariates or zones with <50 observations.

3. Statistical Methods

Having constructed episode-level observations with comprehensive tactical covariates (Section 2), we now detail the statistical framework for estimating mortality risk. Our analysis employs Cox proportional hazards models to quantify how player state, spatial positioning, utility coverage, and numerical advantage dynamically influence survival probability during post-plant retakes. The Cox

framework offers several advantages for this application: it handles right-censored data naturally (the majority of episodes end in zone transitions rather than death), accommodates time-varying covariates through the counting process formulation (Section 2.4), and provides interpretable hazard ratios that directly quantify tactical risk factors.

We estimate separate models for Counter-Terrorists and Terrorists on each bombsite (four configurations total), applying L2 (Ridge) penalization with λ selected via grid search over [0.01, 0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 5.0] to mitigate overfitting in the high-dimensional covariate space (62-120 features per model). Optimal penalty values range from $\lambda = 0.01$ to $\lambda = 0.5$ depending on model complexity and feature count. Model performance is evaluated via 10-fold cross-validation with retake-level splits to prevent data leakage: all episodes from a given retake scenario are assigned to the same fold, ensuring that training and test sets contain episodes from distinct matches. We validate model assumptions through concordance analysis and proportional hazards tests, and employ robust variance estimation (Lin & Wei, 1989) to account for within-retake correlation among episodes.

3.1. Model Specification and Interpretation

We employ Cox proportional hazards regression to model episode-level mortality risk as a function of tactical covariates. The Cox model estimates the hazard function—the instantaneous risk of death at time t given covariate vector X —through a semi-parametric specification:

$$h(t|X) = h_0(t) \cdot \exp\left(\sum_i \beta_i X_i\right) \quad (2)$$

Where $h_0(t)$ is the baseline hazard function (unspecified, estimated non-parametrically) and $\exp(\beta_i)$ represents the hazard ratio associated with covariate X_i . This formulation separates the baseline risk profile—how hazard evolves over episode duration—from the multiplicative effects of tactical covariates. The proportional hazards assumption requires that covariate effects remain constant over time within episodes: a player with 50 HP faces consistently higher hazard than one with 100 HP throughout the episode, with the hazard ratio $\exp(\beta_{HP})$ remaining stable.

Parameter estimation proceeds via partial likelihood [5], which conditions on the set of individuals at risk at each observed death time. For death occurring at time t_i to individual i , the partial likelihood contribution is:

$$L_i(\beta) = \left(\frac{\exp(\beta^T X_i)}{\sum_{k \in R(t_i)} \exp(\beta^T X_k)} \right) \quad (3)$$

where $R(t_i)$ denotes the risk set—all individuals still alive (uncensored) at time t_i . Censored observations contribute to risk sets until their censoring time, then exit without contributing a death event. Maximum partial likelihood estimates $\hat{\beta}$ are obtained via Newton–Raphson optimization, implemented in the Python lifelines library [6].

3.2. Model Specification and Interpretation

We estimate separate Cox models for each combination of team (CT/T) and bombsite (A/B), yielding four configurations that account for asymmetric tactical dynamics. Counter-Terrorist models capture offensive retake behavior, while Terrorist models reflect defensive positioning post plant. Bombsite-specific models allow zone indicator variables and spatial covariates (visibility, connectivity) to exhibit site-dependent effects without imposing cross-site constraints.

We fit three nested model specifications for each team/bombsite combination to assess the incremental value of tactical covariates. Model 1 (baseline) includes only zone indicators, establishing position-specific risk without equipment or tactical context. Model 2 adds player resource covariates (health, armor, helmet, weapon type, and for CT models, defuse kit ownership), testing whether player state improves prediction beyond mere position. Model 3 (full model) builds on Model 2 by incorporating team composition features (teammate equipment quality, positions, and tactical state), utility exposure (smoke, molotov, flashbang effects), numerical advantage, crossfire support, and theoretically-motivated zone-based interactions. We compare models via concordance index to evaluate which covariate classes contribute meaningful predictive power.

Covariate effects are interpreted through hazard ratios (HR): $HR_j = \exp(\beta_j)$ represents the multiplicative change in instantaneous mortality risk for a one-unit increase in covariate X_j , holding other variables constant. For continuous covariates, $HR > 1$ indicates increased hazard (higher risk), while $HR < 1$ indicates protective effect (lower risk). For example, a health covariate with $\beta_{HP} = -0.02$ yields $HR = \exp(-0.02) \approx 0.98$: each additional health point reduces mortality hazard by approximately 2% (since $1 - 0.98 = 0.02$). For dummy variables, a zone indicator with $\beta_{zone} = 0.5$ yields $HR = \exp(0.5) \approx 1.65$, meaning players in that zone face 65% higher mortality risk compared to the reference zone. Categorical variables (weapon types, zone indicators) use reference category encoding, with hazard ratios representing risk relative to the baseline category.

Interaction terms capture conditional effects where the impact of one covariate depends on another. We include theoretically-motivated two-way interactions: zone $\times$ teammate crossfire support tests whether certain zones benefit more from line-of-sight coverage; zone $\times$ smoke exposure interactions capture whether smoke effects vary by position; zone $\times$ molotov exposure tests positional differences in incendiary vulnerability; and zone $\times$ numerical advantage examines whether player count advantage matters more in certain tactical positions. Interactions are constructed only if they occur in more than 3% of episodes to avoid extreme sparsity, though all

constructed terms are retained in the Ridge model with shrinkage proportional to their predictive contribution.

Given the high-dimensional covariate space relative to event counts we apply L2 (Ridge) penalization to stabilize coefficient estimates and mitigate overfitting. The penalty parameter λ is selected via grid search over [0.01, 0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 5.0] using 10-fold cross-validation with retake-level grouping (using sklearn's GroupKFold), ensuring that all episodes from the same retake appear in the same fold to prevent information leakage. Optimal penalty values range from $\lambda = 0.01$ to $\lambda = 0.5$ depending on model complexity and feature count. Ridge regression shrinks coefficients toward zero proportionally while retaining all predictors, yielding regularized estimates that balance bias and variance. The models retain all covariates across specifications: 62 (CT A-site zone model), 44 (CT B-site zone), 80 (T A-site zone), 56 (T B-site zone), 71 (CT A equipment), 52 (CT B equipment), 87 (T A equipment), 63 (T B equipment), 104 (CT A full), 87 (CT B full), 120 (T A full), and 96 (T B full). This yields effective EPV ratios of 6.6 –19.0 across model specifications. While some full models fall below the conventional threshold of 10 events per variable, Ridge shrinkage compensates for the lower EPV, as evidenced by minimal overfitting (typical cross-validation gaps of 0.01–0.03 concordance points).

3.2. Robust Variance Estimation

We estimate separate Cox models for each combination of team (CT/T) and bombsite (A/B), yielding four configurations that account for asymmetric tactical dynamics. Counter-Terrorist models capture offensive retake behavior, while Terrorist models reflect defensive positioning post plant. Bombsite-specific models allow zone indicator variables and spatial covariates (visibility, connectivity) to exhibit site-dependent effects without imposing cross-site constraints.

Standard Cox regression assumes independent observations, but episodes within the same retake share common tactical context—the same player positions, equipment states, and numerical advantage at plant time. This induces correlation: if one CT player dies quickly on a particular retake, their teammates may face elevated risk due to shared disadvantageous positioning rather than independent random variation.

To account for this within-retake correlation, we employ clustered robust variance estimation (sandwich estimator) [7], specifying retake identifiers as the clustering variable. This approach adjusts standard errors and confidence intervals to reflect the grouped data structure, yielding valid inference despite correlated observations. Clustered standard errors are typically wider than naive standard errors, producing conservative p-values and appropriately accounting for the reduced effective sample size when episodes cluster within retakes.

4. Results

We present findings from Cox proportional hazards models estimated on 65,688 post-plant episodes spanning 3,508 professional retake scenarios. After partitioning by team and bombsite, we analyze 21,260 CT episodes (1,713 deaths, 8.0% event rate) and 44,328 T episodes (2,236 deaths, 5.0% event rate). Models achieve strong discriminative performance with concordance indices ranging from 0.607 to 0.806, indicating that tactical covariates substantially outperform random prediction (C-index = 0.50).

4.1. Model Comparison and Predictive Performance

Table 2 summarizes model performance across all twelve specifications (three model types × four team-site combinations). The zone-only baseline models achieve C-indices of 0.607-0.659, confirming that spatial positioning alone provides meaningful mortality prediction. Adding equipment covariates (Model 2) improves performance to 0.692-0.734, with gains of 6.0-11.2 percentage points, demonstrating that player state (health, armor, weapons) contributes predictive value beyond mere position. The full models incorporating tactical coordination features achieve 0.748-0.806, with incremental gains of 3.9-8.7 points over equipment models.

Table 2: Cox Proportional Hazards Model Performance

Model	Team-Site	Episodes	Deaths	EPV	Covariates	C-index	CV C-index	Δ from prior
Zone Only	CT A	15,336	1,137	18.3	62	0.608	0.574	—
	CT B	6,024	576	13.1	44	0.659	0.626	—
	T A	26,882	1,522	19.0	80	0.631	0.611	—
	T B	17,446	714	12.8	56	0.607	0.540	—
Equipment	CT A	15,336	1,137	16.0	71	0.692	0.663	+8.2
	CT B	6,024	576	11.1	52	0.734	0.703	+7.6
	T A	26,882	1,522	17.5	87	0.694	0.677	+5.5
	T B	17,446	714	11.3	63	0.719	0.688	+13.6
Full Model	CT A	15,336	1,137	10.9	104	0.748	0.718	+4.9
	CT B	6,024	576	6.6	87	0.773	0.734	+2.8
	T A	26,882	1,522	12.7	120	0.789	0.775	+9.8
	T B	17,446	714	7.4	96	0.806	0.786	+9.9

Note: C-index = concordance index (discrimination); CV C-index = 10-fold cross-validated concordance; EPV = events per variable (deaths/covariates); Δ = percentage point improvement from previous model tier. All models use Ridge (L2) regularization with penalty λ selected via grid search (optimal values range 0.01–0.5) and clustered standard errors by retake ID.

4.2. Survival Patterns

These performance improvements have direct practical implications for competitive play. The progression from baseline zone models (C-index 0.607-0.659) to full models incorporating crossfire geometry and teammate support (C-index 0.748-0.806) represents a shift from marginal predictive power to actionable tactical intelligence. A C-index of 0.806 means the model correctly ranks survival times for over 80% of player pairs—sufficient to distinguish high-risk from low-risk setups in real-time decision making. Critically, the majority of this discriminative power emerges only when geometric relationships are included: equipment adds 6-11 percentage points over position alone, but crossfire support, empirically-derived sightlines, and teammate positioning add another 4-10 points. This validates that post-plant survival is fundamentally a team coordination problem, not an individual skill problem. Holding the "correct" position matters, but holding positions that create crossfire angles and control empirically-validated sightlines matters more. The fact that we can predict survival with up to 81% accuracy using only initial positioning geometry—before any shots are fired—demonstrates that tactical setup, not mechanical execution, dominates post-plant outcomes in professional Counter-Strike.

While the Cox proportional hazards models establish which factors predict survival, the real value lies in using survival analysis to validate tactical assumptions from professional play. By stratifying Kaplan-Meier curves across strategic variables—crossfire positioning, smoke placements, and site-specific zones—we can statistically test whether common in-game decisions actually improve survival outcomes. This approach transforms anecdotal tactical knowledge into empirically validated strategy, revealing not only which positions are safest, but whether widely-used utility setups provide measurable survival advantages or, counterintuitively, increase risk.

4.2.1. Safest Post Plant Positions with Crossfire

Table 3 presents the five safest positions with crossfire support for each bomb site, illustrating how tactical geometry compounds with individual positioning to maximize survival. After adjusting for bomb plant location, health, armor, weapon type, utility usage, numerical advantage, and teammate composition, positional effects remain substantial and statistically significant (all $p < 0.001$ for A-site top 5, $p < 0.005$ for B-site). On A-site, positions offering cover from primary Counter-Terrorist approach angles—Before A-Main (HR=0.795 with crossfire, 20.5% risk reduction), Palace Exit (HR=0.805, 19.5%), and Mid Ramp (HR=0.824, 17.6%)—provide the strongest protection when coupled with teammate support. The multiplicative nature of crossfire is evident: a player in Before A-Main benefits from inherent positional safety (solo HR=0.819), but adding a teammate positioned to engage threats from a different angle amplifies this advantage to a combined 20.5% risk reduction. B-site exhibits similar patterns, with B Apps First Section (HR=0.858, 14.2% reduction) and B Pillar 1 (HR=0.883, 11.7% reduction) emerging as optimal holdpoints. Notably, these protective effects persist even after controlling for all equipment and team composition variables, confirming that spatial configuration and crossfire geometry—not superior loadouts or numerical advantage—drive positional strength. The consistent significance across both sites ($p < 0.001$ to $p < 0.005$) demonstrates that empirically-derived optimal positions are robust and generalizable tactical principles, not site-specific artifacts.

Table 3: Top 5 Safest Positions with Crossfire Support

A-Site (Terrorist)					
Position	Zone	Solo HR	With Crossfire HR	Risk Reduction	p-value
Before A-Main	64	0.819	0.795	20.5%	<0.001
Palace Exit	73	0.829	0.805	19.5%	<0.001
Mid Ramp	115	0.849	0.824	17.6%	<0.001
Underpass Close	118	0.856	0.831	16.9%	<0.001
A Ramp Room	63	0.857	0.832	16.8%	<0.001

B-Site (Terrorist)					
Position	Zone	Solo HR	With Crossfire HR	Risk Reduction	p-value
B Apps First Section	262	0.880	0.858	14.2%	<0.001
B Pillar 1	250	0.906	0.883	11.7%	<0.001
Bench Corner	239	0.919	0.896	10.4%	<0.005
Cat Brick	139	0.919	0.896	10.4%	<0.005
Ladder Room	137	0.934	0.910	9.0%	<0.005

Note: Hazard ratios adjusted for bomb plant position, health, armor, weapon type, utility usage, numerical advantage, and teammate composition. Crossfire support is defined as having a teammate positioned to engage enemy threat zones from a different angle (empirically derived from visibility graph). General crossfire effect: A-site HR=0.971 ($p<0.001$), B-site HR=0.975 ($p<0.001$).

4.2.2. Best Smokes to Throw Post Plant

Figure 8 represents which smokes on average are best to throw in post-plant situations to help increase survivability. The analysis reveals a counterintuitive finding: not all smokes help the team that throws them.

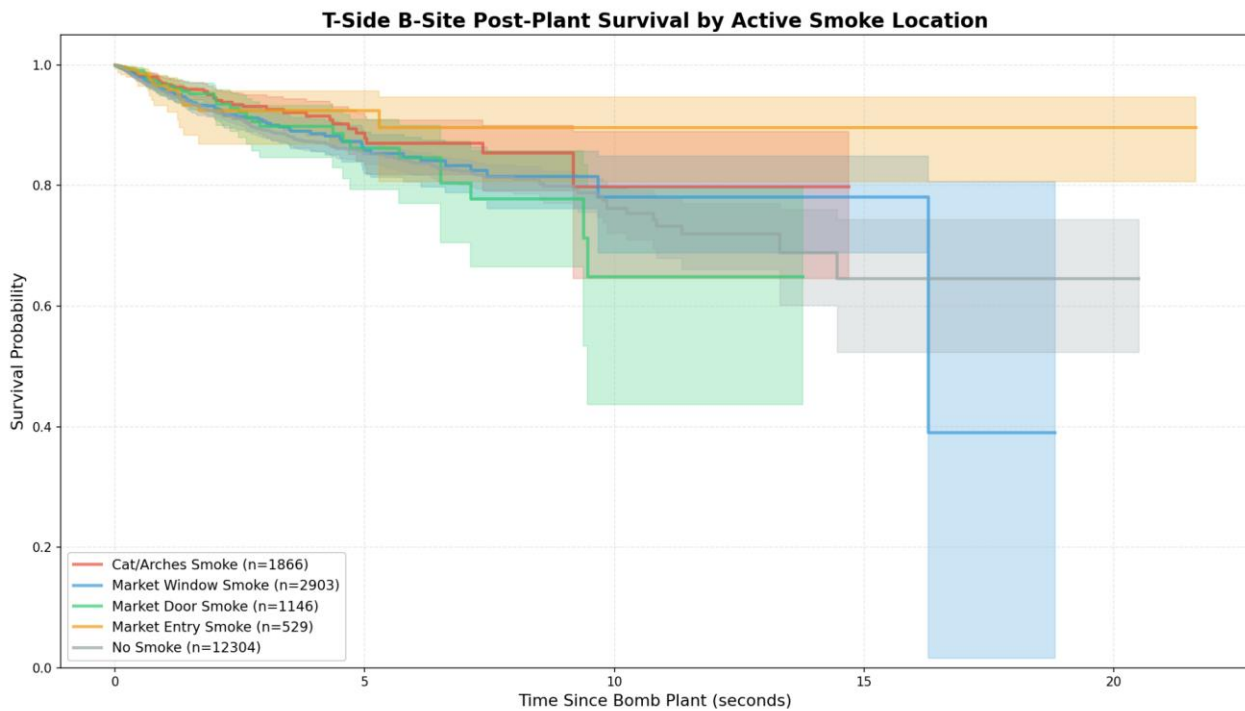
On A-site, the best smoke to throw post-plant is Jungle smoke ($p<0.001$), which allows T-side players to take control of space and get into power positions such as Jungle/Connector Cubby (Zone 47). The survival curves show throwing Stairs smoke or rethrowing it statistically isn't the best to throw in a post-plant—players with Stairs smoke active show worse survival than those with no

smoke at all (median 11.97s vs 20.46s). This occurs because committing a smoke to stairs opens up other areas like Jungle or Ticket that are better for CTs to push from, leaving those critical retake paths uncontested.

On B-site, getting Market Entry (Zone 170) smoked off has a very high survival rate, once again showing the importance of taking control of space for the T-side post-plant. Cat/Arches smoke also demonstrates strong survival outcomes, while Market Window smoke remains the most frequently used (n=2903) despite showing only marginal improvement. These patterns demonstrate that effective smoke placement depends critically on understanding retake pathways—smokes that block the most common and dangerous CT approaches provide measurable survival benefits, while misallocating utility to less impactful positions leaves more dangerous areas exposed.



(a)



(b)

Figure 8: Kaplan-Meier survival curves comparing post-plant smoke locations for T-side on A-site (a) and B-site (b). Curves show survival probability over time since bomb plant, with 95% confidence intervals. Only Jungle smoke on A-site demonstrates statistically significant survival improvement ($p < 0.001$), while Stairs smoke shows worse outcomes than no smoke. On B-site, Market Entry smoke exhibits the highest survival rates, suggesting spatial control of retake pathways is more critical than visual obstruction.

4.2.3. The Clutch: Crossfire Support Under Pressure

Figure 9 reveals how crossfire effectiveness scales with numerical advantage, exposing the tactical mechanics behind successful clutch situations. Across all three scenarios, crossfire support (having at least one teammate) provides statistically significant survival benefits, but the magnitude varies dramatically based on numerical context.

When advantaged (≥ 1 extra player), crossfire support nearly doubles median survival time (4.14s without vs 9.10s with crossfire), demonstrating that numerical superiority becomes decisive when combined with tactical coordination. The large gap between survival curves shows that teams with both numbers and crossfire can systematically close out rounds.

When even (equal players), crossfire provides modest but meaningful improvement (10.78s → 11.97s median survival). The converging survival curves suggest that in fair fights, positioning and coordination offer incremental advantages rather than overwhelming ones.

When disadvantaged (outnumbered), the graph shows that if you don't have crossfire support and you're disadvantaged, you probably won't make it. However, this can also be interpreted in clutch situations: if a player is able to find isolated fights and take them, their percentage chance of winning the round increases dramatically. Avoiding crossfires turns numerical disadvantage into a series of winnable 1v1s rather than coordinated executions.

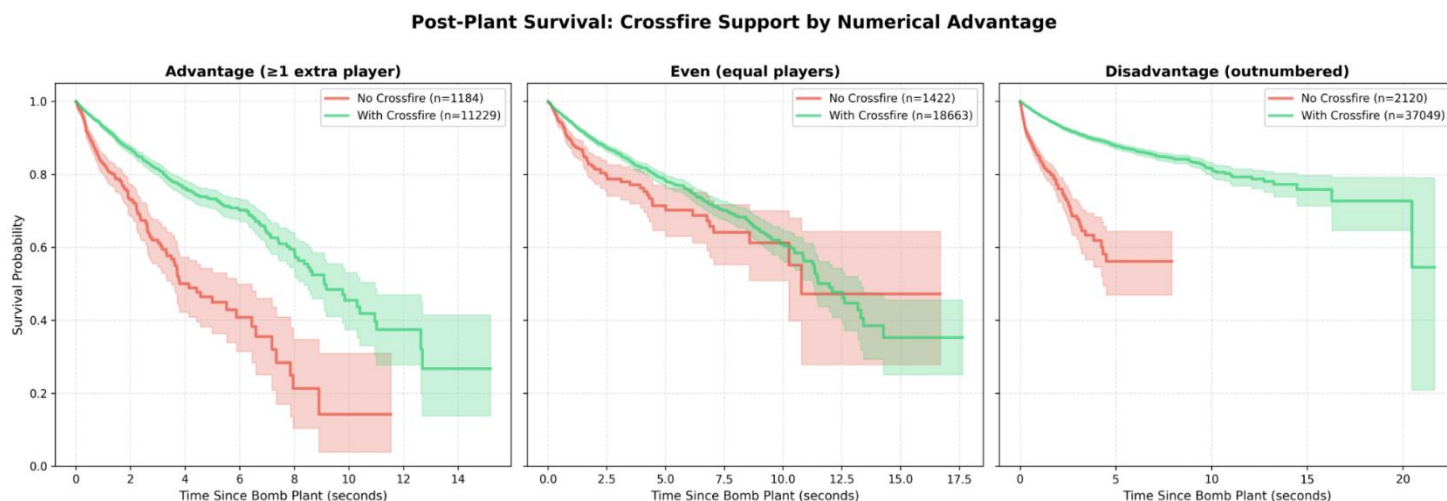


Figure 9: Kaplan-Meier survival curves comparing crossfire support across numerical advantage scenarios (advantaged, even, disadvantaged), adjusted for site, health, armor, weapons, and utility.

4.2.4. Case Study

Using the coefficients from our trained Cox model, we can analyze hazard rate changes in real professional matches, providing tactical insights into round dynamics. Figure 10 examines a 2v1 post-plant scenario from fnatic vs Imperial on Mirage (Round 9, 2024 FiReLEAGUE Global Finals), where Imperial's T-side players HEN1 and decency defend a bomb planted at A Default against a single CT retake attempt.

The hazard rate timeline (Figure 10) illustrates how the model-predicted risk of death changes as the round progresses through four distinct phases. At bomb plant ((1), t=0.00s), Imperial holds a 2v1 numerical advantage with HEN1 positioned at A_Default (Zone 30, AWP, 100 HP, Full Armor) and decency at Triple_Conn_Side (Zone 22, AK-47, 100 HP, Full Armor). A Stairs smoke remains active with 7.5 seconds remaining from the pre-plant execution. The initial hazard rate is 0.978, calculated from the model coefficients: Zone HR (1.06) × Numerical advantage (0.94) × Teammates (0.99) × Smoke (1.00). Notably, this represents only a 2.2% risk reduction despite the 2v1 advantage, as the players were positioned in separate zones without an established crossfire setup. Furthermore, HEN1's position at A_Default (Zone 30)—the bomb plant location—was particularly

vulnerable with no Ticket smoke active, leaving him exposed to early aggression from Ticket Entry (Zone 9).

At 0.93 seconds (2), the CT pushes aggressively from Ticket Entry (Zone 9) and immediately eliminates player HEN1 at A Default (Zone 30), dealing 141 damage. This early death validates the model's risk assessment: despite the numerical advantage, HEN1's exposed position at the bomb site made him highly vulnerable. The round transitions to a 1v1, with teammate decency at Triple Conn Side (Zone 22) now isolated without teammate support. The hazard rate increases to 1.057, representing an 8% increase. Critically, the hazard rate does not skyrocket despite the loss of numerical advantage because the model accurately captures the CT's disadvantageous position—pushing from Ticket Entry (Zone 9) with no utility to throw, no active smokes to support the push, many angles to clear, and zero information about defender positions. The model correctly assesses that even in a 1v1, decency retains reasonable survival odds given the CT's tactical constraints.

At 3.50 seconds (3), decency at Triple Conn Side secures the trade kill, eliminating the CT attacker. Critically, decency took zero damage throughout the entire round, demonstrating how perfect positioning enabled a safe trade despite being caught in a 1v1. With no enemy combatants remaining, the hazard rate drops 91% to 0.100, reflecting only the minimal threat of failing to clear the bomb detonation radius.

At 8.48 seconds (4), the bomb detonates with decency safely clear of the blast radius, securing the round for Imperial. The hazard rate reaches zero as all threats are eliminated. The model-predicted hazard rates align with the round outcome, accurately capturing how risk changed as players were eliminated and tactical situations evolved. This case study demonstrates the model's ability to quantify tactical dynamics in real-time, providing interpretable insights into the factors that influence survival in professional Counter-Strike 2 post-plant scenarios. The model successfully predicted HEN1's vulnerability at A Default and appropriately moderated the hazard increase when the CT pushed from Ticket Entry without utility support, showcasing its nuanced understanding of positional risk and tactical context.

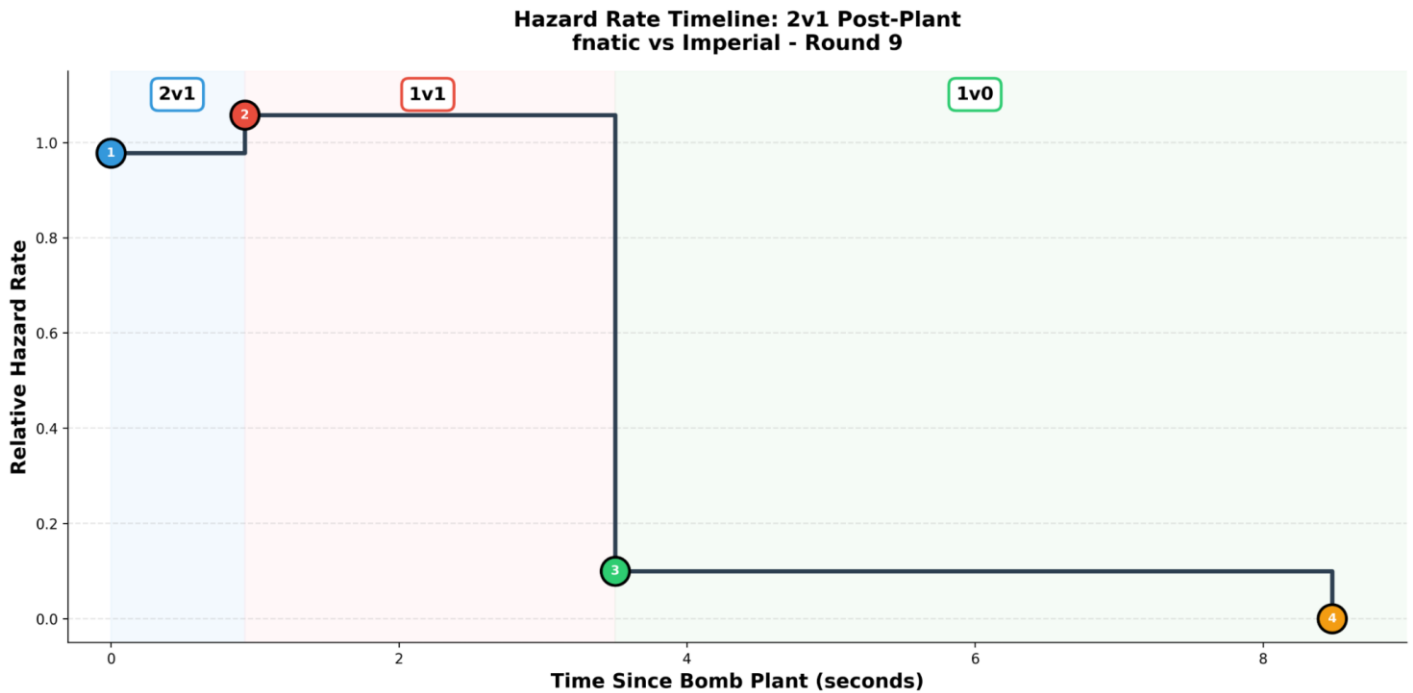


Figure 10: Hazard rate timeline for a 2v1 post-plant scenario (fnatic vs Imperial, Round 9, Mirage A-site). Numbered markers indicate critical tactical transitions: (1) Bomb plant establishes 2v1 T advantage with low baseline mortality risk; (2) HEN1 at A Default (Zone 30) eliminated by CT push from Ticket Entry (Zone 9), reducing to 1v1 and increasing T death risk; (3) decency at Triple Conn Side (Zone 22) trades the elimination, establishing 1v0 and dropping mortality risk near zero; (4) Round complete, Imperial secures the win. Color-coded background regions denote numerical advantage phases (2v1, 1v1, 1v0). Hazard rates estimated using the T A-site full model with grid-search optimized Ridge regularization ($\lambda=0.01$, C-index=0.789).

5. Discussion

5.1. Model Performance and Tactical Complexity

Our Cox proportional hazards models achieve concordance indices ranging from 0.748 to 0.806 across all scenarios, with T-side post-plant models (0.789-0.806) outperforming CT-side retake models (0.748-0.773). This performance gap reflects fundamental asymmetries in Counter-Strike 2's post-plant phase: defenders know bomb locations and can prepare positions, while attackers must gather information through risky map control. The strong T-side performance validates that defensive positioning follows predictable risk patterns when players have positional certainty, while CT retake uncertainty—not knowing defender positions or utility setups—introduces irreducible variance that limits prediction accuracy.

The model's ability to achieve up to 81% discrimination suggests that spatial positioning, utility deployment, and teammate coordination explain much of the variance in professional survival outcomes. However, the remaining ~19% unexplained variance points to factors our current covariates cannot capture: individual mechanical skill, psychological pressure in clutch situations, and most critically, information asymmetries about enemy locations.

5.2. Challenging Conventional Wisdom

Several findings challenge common tactical assumptions in professional play. The Stairs smoke (Zone 44) increases defender hazard rates despite being a "standard" A-site setup, revealing a tactical miscalculation: while it blocks CT vision from Ticket Entry, it also obscures defenders' ability to spot incoming pushes. Moreover, defenders may not anticipate CT players sitting inside or behind the smoke, waiting for it to fade before engaging—a timing exploit that turns the defensive utility against its users. This demonstrates that utility value is context-dependent and dynamic—smokes are not universally protective, and teams must consider both what they block from enemies, what they block for defenders, and how opponents might exploit the temporary vision denial.

Similarly, popular aggressive positions like Ropz (Zone 14, HR=1.055) and Jungle Cubby (Zone 153, HR=0.930) show contrasting risk profiles despite their prevalence in professional play. A Ropz increases individual death risk by 5.5%, while Jungle Cubby reduces it by 7%, yet both positions serve the same tactical purpose: taking aggressive map control in post-plant scenarios to control connector approaches and force isolated 1v1 engagements rather than facing coordinated retakes. This demonstrates that optimal team strategy sometimes requires specific positioning choices that balance individual survival with round-winning map control.

The modest coefficients for numerical advantage (HR=0.938) and crossfire support (HR=0.971) initially seem surprising—shouldn't 2v1 situations be far safer? Our case study explains this: coefficients measure average effects across all positioning configurations, including poorly coordinated setups. When players position correctly (Figure 9), crossfire with numerical advantage nearly doubles survival time. The model's aggregate coefficients are modest because many teams fail to translate numerical superiority into tactical crossfire setups.

5.3. The Critical Gap: Information State

The primary limitation of our model—and the most important direction for future work—is quantifying information asymmetry. In the case study, the model predicted HR=0.978 for the 2v1 advantage, but this assumes equal information states. In reality, the CT had zero information about defender positions while defenders heard the Ticket Entry push. The model cannot distinguish between "2v1 with full information" (very safe) and "2v1 with no information about enemy locations" (moderately safe), collapsing both into a single coefficient.

We are developing methods to incorporate information state through audio event tracking (footsteps, gunfire, utility sounds), line-of-sight analysis using collision mesh geometry, and information decay models that account for how positional knowledge degrades over time. Preliminary work suggests information state may explain much of the currently unexplained variance in CT retake models, where attackers systematically lack defender position knowledge.

5.4. Practical Impact and Conclusion

This research provides coaches and analysts with quantitative tools to evaluate positioning decisions objectively. Rather than relying on intuition or anecdotal evidence, teams can now calculate expected survival benefits of specific setups, identify which utility deployments provide measurable value, and optimize crossfire configurations based on numerical context. The real-time hazard analysis framework enables post-match review where specific positioning choices can be evaluated against model predictions, transforming subjective tactical discussions into data-driven optimization problems.

By analyzing episodes from elite matches, we identified specific positions and tactical configurations that reduce death risk by up to 20.5% when combined with teammate support. Our case study validation demonstrates the model accurately predicts real-time hazard changes in professional matches, confirming it captures genuine tactical dynamics rather than spurious correlations. The findings challenge conventional tactical assumptions—revealing that "standard" utility setups can increase risk when they obscure defender vision, and that popular aggressive positions accept elevated individual risk to maximize team win probability.

Most critically, we identified information asymmetry as the key limitation preventing higher prediction accuracy, motivating ongoing development of audio tracking and line-of-sight analysis to quantify what players know about enemy positions. All code, data preprocessing pipelines, and trained models are publicly available on GitHub [8], enabling replication and extension of this work. Beyond immediate coaching applications, this methodology demonstrates how survival analysis can extract actionable insights from esports gameplay data, providing a template for analysis in other tactical shooters like Valorant or Rainbow Six Siege. By transforming subjective tactical discussions into quantifiable survival benefits, this research provides an evidence-based foundation for strategic decision-making in professional Counter-Strike 2.

References

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Appendix A – Changes from Submitted Abstract

The final implementation differs from the submitted abstract in two primary ways. First, scope was narrowed to a single map (Mirage) rather than analyzing clutches across all competitive maps. This decision enabled deeper spatial analysis through map-specific zone classification (291 zones), empirically-derived visibility graphs, and connectivity networks that would not generalize across different map geometries. Focusing on Mirage—the most-played professional map—ensured sufficient sample density while allowing for granular positional covariates (zone indicators, plant spots, crossfire metrics) that capture tactical nuance. Second, the evaluation framework shifted from clutch win prediction to post-plant survival analysis. Rather than modeling binary clutch outcomes (1vX scenarios where the solo player must eliminate all opponents), the final study analyzes episode-level mortality risk for all players during post-plant phases. This broader framing uses Cox proportional hazards models with concordance index as the primary performance metric, rather than Brier scores and AUC for binary classification. These changes maintained the core research question—quantifying how spatial-temporal geometry influences survival probability—while focusing the analysis on a setting where positional strategy has maximal competitive impact: defending or retaking a planted bomb on Mirage.

Appendix B – Zone Connectivity and Callout Location Reference

Zone ID	Location Name	JSON Identifier
1	T Spawn	T_Spawn
2	CT Spawn	CT_Spawn
3	Deep CT Spawn	Deep_CT_Spawn
4	CT Exit	CT_Exit
5	Blue Door	Blue_Door
6	CT Ramp Lower	CT_Ramp_Lower
7	Ct Ramp Upper	CT_Ramp_Upper
8	Trash Can	Trash_Can
9	Ticket Entry	Ticket_Entry
10	Deep Ticket	Deep_Ticket
11	Ticket	Ticket
12	Ticket Box	Ticket_Box
13	Top Ticket	Top_Ticket
14	Ropz	Ropz
15	Ticket Exit	Ticket_Exit
16	Reverse Ticket	Reverse_Ticket
17	Bench Triple Side	Bench_Triple
18	Triple CT Side	Triple_CT_Side
19	Ropz Open	Ropz_Open
20	Ticket Exit Open	Ticket_Exit_Open
21	Door	Door
22	Conn Side Triple	Conn_Side_Triple

23	Top Triple	Top_Triple
24	Main Side Triple	Main_Side_Triple
25	Triple Default	Triple_Default
26	Backsite A-site	Backsite_A
27	Mid A-site	Mid_A
28	CT Side Default	CT_Side_Default
29	Ninja	Ninja
30	Default A	Default_A
31	Firebox/Default	Firebox_Default
32	Firebox	Firebox
33	Open Firebox	Open_Firebox
34	Open Default	Open_Default
35	Under Balc/Default	Under_Balc_Default
36	Open Default	Open_Default_A
37	Open 1A	Open_1A
38	Open 2A	Open_2A
39	Open 3A	Open_3A
40	Open 4A	Open_4A
41	Bench 1A	Bench_1A
42	Bench 2A	Bench_2A
43	Open 5A	Open_5A
44	Stairs Platform	Stairs_Platform
45	Jungle Open 1A	Jungle Open 1A
46	Jungle Exit	Jungle_Exit
47	Connector/Jungle Cubby	Conn_Jungle_Cubby
48	Connector Exit	Connector_Exit
49	Lower Stairs	Lower_Stairs_A
50	Mid Stairs	Mid_Stairs_A
51	Upper Stairs	Upper_Stairs_A
52	Sandwich	Sandwich
53	Sandwich Open	Sandwich_Open
54	Tetris Open	Tetris_Open
55	Open 6A	Open_6A
56	Tetris Cubby	Tetris_Cubby
57	Tetris Tucked	Tetris_Tucked
58	Open 7A	Open_7A
59	Reverse Apex	Reverse_Apex
60	A Main Exit	A_Main_Exit
61	A Main Cubby	A_Main_Cubby
62	A Main	A_Main
63	Pottery	Pottery
64	A Main Stairs	A_Main_Stairs
65	A Main Cubby	A_Main_Cubby
66	A Main Entrance	A_Main_Entrance

67	Entrance Open 1	Entrance_Open_1
68	Entrance Rafters 1	Entrance_Rafters_1
69	Heaven 1	Heaven_1
70	Heaven 1 Cubby	Heaven_1_Cubby
71	Heaven 2	Heaven_2
72	Heaven 2 Cubby	Heaven_2_Cubby
73	Palace Exit	Palace_Exit
74	Palace	Palace
75	Palace Open 1	Palace_Open_1
76	Palace Open 2	Palace_Open_2
77	Palace Open 3	Palace_Open_3
78	Palace Open 4	Palace_Open_4
79	Palace Open 5	Palace_Open_5
80	Palace Open 6	Palace_Open_6
81	Palace Open 7	Palace_Open_7
82	Palace Entry Cubby	Palace_Entry_Cubby
83	Palace Open 8	Palace_Open_8
84	Palace Open 9	Palace_Open_9
85	Palace Entry	Palace_Entry
86	Entrance Open 2	Entrance_Open_2
87	Entrance Rafters 2	Entrance_Rafters_2
88	Entrance Cubby	Entrance_Cubby
89	T Spawn Open 1	Tspawn_Open_1
90	T Spawn Open 2	Tspawn_Open_2
91	T Spawn Open 3	Tspawn_Open_3
92	T Spawn Open 4	Tspawn_Open_4
93	T Spawn Open 5	Tspawn_Open_5
94	Apps Ramp	Apps_Ramp
95	Apps Stairs	Apps_Stairs
96	TV Platform	TV_Platform
97	Cart	Cart
98	Cart Open	Cart_Open
99	Alley Cubby	Alley_Cubby
100	Side Alley	Side_Alley
101	Alley Jump Up	Alley_Jump_Up
102	Top Mid 1	Top_Mid_1
103	Top Mid 2	Top_Mid_2
104	Top Mid 3	Top_Mid_3
105	Top Mid 4	Top_Mid_4
106	Top Mid 5	Top_Mid_5
107	Top Mid 6	Top_Mid_6
108	Top Mid 7	Top_Mid_7
109	Top Mid 9	Top_Mid_9
110	Top Mid Cubby	Top_Mid_Cubby

111	Catwalk 1	Catwalk_1
112	Bottom Mid 1	Bottom_Mid_1
113	Chair Open	Chair_Open
114	Chair	Chair
115	Bottom Mid 2	Bottom_Mid_2
116	Catwalk 2	Catwalk_2
117	Bottom Mid 3	Bottom_Mid_3
118	Bottom Mid 4	Bottom_Mid_4
119	Mid Bench	Mid_Bench
120	Mid Bench Cubby	Mid_Bench_Cubby
121	Bottom Conn	Bottom_Conn
122	Bottom Conn Cubby	Bottom_Conn_Cubby
123	Mid Connector	Mid_Connector
124	Top Conn Cubby 1	Top_Conn_Cubby_1
125	Top Conn	Top_Conn
126	Top Conn Cubby 2	Top_Conn_Cubby_2
127	Bottom Mid 5	Bottom_Mid_5
128	Under Window Cubby	Under_Window_Cubby
129	Underpass Ramp	Underpass_Ramp
130	Underpass Ramp Cubby 1	Underpass_Ramp_Cubby_1
131	Underpass Ramp Cubby 2	Underpass_Ramp_Cubby_2
132	Under Window	Under_Window
133	Ladder Room Vent	Ladder_Room_Vent
134	Window	Window
135	Window Cubby 1	Window_Cubby_1
136	Window Cubby 2	Window_Cubby_2
137	Ladder Room	Ladder_Room
138	Catwalk 3	Catwalk_3
139	Catwalk 4	Catwalk_4
140	Catwalk 5	Catwalk_5
141	Catwalk 6	Catwalk_6
142	Window Room 1	Window_Room_1
143	Window Room Cubby 1	Window_Room_Cubby_1
144	Window Room 2	Window_Room_2
145	Window Room 3	Window_Room_3
146	Window Room 4	Window_Room_4
147	Window Room Cubby 2	Window_Room_Cubby_2
148	Window Room 5	Window_Room_5
149	Jungle Door	Jungle_Door
151	Jungle 1	Jungle_1
152	Jungle 2	Jungle_2
153	Jungle Corner	Jungle_Corner
154	Jungle Cubby	Jungle_Cubby
155	Murder Hole	Murder_Hole

156	Murder Hole Entrance 1	Murder_Hole_Entrance_1
157	Murder Hole Entrance 2	Murder_Hole_Entrance_2
158	Murder Hole Entrance 3	Murder_Hole_Entrance_3
159	Murder Hole Entrance 4	Murder_Hole_Entrance_4
160	Murder Hole Entrance 5	Murder_Hole_Entrance_5
161	Murder Hole Headshot	Murder_Hole_Headshot
162	Market Entry Cubby 1	Market_Entry_Cubby_1
163	CT Spawn Open 1	CT_Spawn_Open_1
164	CT Spawn Open Cubby	CT_Spawn_Open_Cubby
165	CT Spawn Open 2	CT_Spawn_Open_2
166	CT Potts	CT_Potts
167	CT Potts Platform	CT_Potts_Platform
168	Market Entry Cubby 2	Market_Entry_Cubby_2
169	Market Entry	Market_Entry
170	Market 1	Market_1
171	Market 2	Market_2
172	Market 3	Market_3
173	Market 4	Market_4
174	Market 5	Market_5
175	Market Window	Market_Window
176	Market 6	Market_6
177	Freezer Cubby	Freezer_Cubby
178	Market 7	Market_7
179	Market 8	Market_8
180	Market 9	Market_9
181	Market Exit Cubby	Market_Exit_Cubby
182	Market Exit	Market_Exit
183	Open 1B	Open_1B
184	Dumpster	Dumpster
185	Open 2B	Open_2B
186	Market Window Exit	Market_Window_Exit
187	Open 3B	Open_3B
188	Open 4B	Open_4B
189	Open Cubby 1	Open_Cubby_1
190	Open 5B	Open_5B
191	Empty Open	Empty_Open
192	Open 6B	Open_6B
193	Open 7B	Open_7B
194	Apps Headshot	Apps_Headshot
195	Pillar 4	Pillar_4
196	Open 8B	Open_8B
197	Open Cubby 2	Open_Cubby_2
198	Open 9B	Open_9B
199	Cat Plant	Cat_Plant

200	Open 10B	Open_10B
201	Open 11B	Open_11B
202	Open 12B	Open_12B
203	Open 13B	Open_13B
204	Open 14B	Open_14B
205	Arches 2 Open	Arches_2_Open
206	Arches 2 Cubby	Arches_2_Cubby
207	Arches 2 Tucked	Arches_2_Tucked
208	Catwalk 7	Catwalk_7
209	Catwalk 8	Catwalk_8
210	Middle Arches	Middle_Arches
211	Arches 1 Deep	Arches_1_Deep
212	Arches 1 Open	Arches_1_Open
213	Get Right	Get_Right
214	Arches 1 Tucked	Arches_1_Tucked
215	Open 15B	Open_15B
216	Open 16B	Open_16B
217	Open 18B	Open_18B
218	B-Site 1	B_Site_1
219	B-Site 2	B_Site_2
220	Open 19B	Open_19B
221	Open 20B	Open_20B
222	Open 21B	Open_21B
223	Under Window B	Under_Window_B
224	Pillar 2	Pillar_2
225	Pillar 2 Open	Pillar_2_Open
226	Under Balcony Open	Under_Balcony_Open
227	Open 22B	Open_22B
228	Default B	Default_B
229	Open 23B	Open_23B
230	B-Site 3	B_Site_3
231	B-Site 4	B_Site_4
232	Safe Plant	Safe_Plant
233	B-Site 5	B_Site_5
234	B-Site 6	B_Site_6
235	Open 24B	Open_24B
236	B-Site 7	B_Site_7
237	Open 24B	Open_24B
238	Open 25B	Open_25B
239	Bench Close Angle	Bench_Close_Angle
240	Bench Deep 1	Bench_Deep_1
241	Bench Deep 2	Bench_Deep_2
242	Bench Deep 3	Bench_Deep_3
243	Bench Deep 4	Bench_Deep_4

244	Jail Open	Jail_Open
245	Jail	Jail
246	Bench B	Bench_B
247	Reverse Bench	Reverse_Bench
248	Open 26B	Open_26B
249	Open 27B	Open_27B
250	Pillar 1	Pillar_1
251	Open 28B	Open_28B
252	Open 29B	Open_29B
253	Open Van	Open_Van
254	Top Van	Top_Van
255	Under Balcony Open	Under_Balcony_Open
256	Under Balcony Tucked	Under_Balcony_Tucked
257	Van Tucked	Van_Tucked
258	Van Box Boost	Van_Box_Boost
259	Van Jump Up	Van_Jump_Up
260	Balcony Platform	Balcony_Platform
261	Balcony Cubby	Balcony_Cubby
262	Apps End	Apps_End
263	Apps 2	Apps_2
264	Apps 3	Apps_3
265	Apps 4	Apps_4
266	Apps Cubby	Apps_Cubby
267	Apps 5	Apps_5
268	Apps 6	Apps_6
269	Apps 7	Apps_7
270	Apps 8	Apps_8
271	Apps 9	Apps_9
272	Apps Kitchen	Apps_Kitchen
273	Apps Entrance 1	Apps_Entrance_1
274	Apps Entrance 2	Apps_Entrance_2
275	Underpass Stairs 2	Underpass_Stairs_2
276	Underpass Stairs 1	Underpass_Stairs_1
277	Back Alley 1	Back_Alley_1
278	Back Alley 2	Back_Alley_2
279	Back Alley 3	Back_Alley_3
280	Back Alley 4	Back_Alley_4
281	House	House
282	TV Room	TV_Room
283	Underpass 1	Underpass_1
284	Underpass 2	Underpass_2
285	Underpass 3	Underpass_3
286	Under Balcony 1	Under_Balcony_1
287	Under Balcony Tucked	Under_Balcony_Tucked

288	Under Balcony 2	Under_Balcony_2
289	Under Balcony 3	Under_Balcony_3
290	Under Balcony 4	Under_Balcony_4
291	Under Balcony 5	Under_Balcony_5

